1. For each statement (a)–(d), choose the most appropriate property of a complex system from the list below. Write the property in the space provided.

   - Adaptation
   - Emergence
   - Heterogeneity
   - Self-organisation

[4 marks]

(a) A flock of birds spontaneously arrange themselves into a structured formation without any central controller. _____

(b) A traffic jam forms through the interactions of individual drivers, even though no driver intends to create it. _____

(c) Drivers in a traffic system have different behaviours and characteristics. _____

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2. What is the difference between a computational model and a simulation?

A computational model is the mathematical or algorithmic representation of a real-world system or process, expressed in a way that a computer can work with. (1) A simulation is the execution of a computational model (1)

3. Consider a connected undirected graph in which each node has degree $k_i = 2$ and local clustering coefficient $C_i = 1$. Sketch the graph and determine the number of nodes n. [2 marks]

network sketch (1)
n = 3 (1)

4. Erdős-Rényi random graphs can be generated using G(n,p) or G(n,m). Explain how edges are generated in each model and state the main difference between the two models. [3 marks]

(1) for each graph how edges are generated and (1) for difference.
In G(n,p), the graph has n nodes and each possible edge is included independently with probability p. Therefore, the number of edges can vary between different graph realisations. (1)

In G(n,m), the graph has n nodes and exactly m edges are selected randomly from all possible edges. Therefore, the number of edges is fixed at m. (1)
The main difference is that G(n,p) fixes the probability of an edge but not the total number of edges, whereas G(n,m) fixes the total number of edges but randomises which edges are present. (1)

"Clustering coefficient quantifies the likelihood that two nodes that are connected to the same node are also connected to each other. (A friend of my friend is my friend)".

Fill in the body of `node_clustering(G,u)` (with python) so that it computes the local clustering coefficient.

You may choose to use:

(a) `all_pairs(g)` to generate all pairs of nodes from the vertex set g

(b) `G.has_edge(u,v)` to establish whether an edge between two nodes u and v exists.

You do not need to define these functions if you use them. [8 marks]

```
def node_clustering(G, u):
    """Computes local clustering coefficient for `u`.

    G: Graph
    u: node

    returns: float
    """
    # FILL THIS IN
```

```
def node_clustering(G, u):
    """Computes local clustering coefficient for `u`.

    G: Graph
    u: node

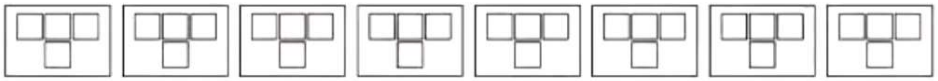
    returns: float
    """
    neighbors = G[u]
    k = len(neighbors)
    if k < 2: #no triangle can form if you have less than two neighbours
        return np.nan #nan means not a number

    possible = k * (k-1) / 2
    exist = 0
    for v, w in all_pairs(neighbors):
        if G.has_edge(v, w):
            exist +=1
    return exist / possible
```

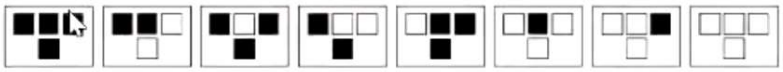
Suggested marking scheme:
(1) for finding neighbours of node u
(1) for ensuring sufficient neighbours for triangle to form and (1) returning nan if clustering cannot be calculated
(1) for determining possible edges
(3) for counting edges that actually exist (1 mark for idea and 2 for a proper implementation)
(1) for returning proportion of edges that exist

6. Consider a one-dimensional cellular automaton in which each cell updates according to Wolfram's Rule 184. Each cell has two possible states one is true (1, black, on etc), the other is false (0, white, off etc). Assume cells outside the shown lattice are in state 0/white.

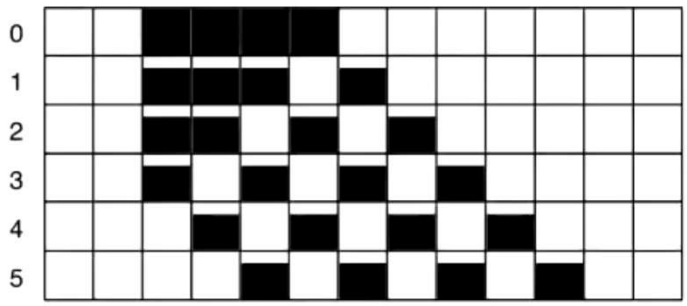
   (a) There are 8 possible configurations of a 3-celled patch (top row). What are they and what are their outputs (bottom row) according to Rule 184. Show your answer by coloring the 1s (black/on) and leaving 0s blank (white/off) with configurations ordered from left to right as largest to smallest binary values (exactly as Wolfram would have). [2 marks]



0.25 per configuration. Top and bottom, both should be correct.



1 for each correct generation. No partial marks.



(a) What are Methuselahs? [2 marks]

Methuselahs are simple starting patterns (1) that are (surprisingly) long-lived (1). For example, the r-pentomino reaches a stable configuration after more than 1000 steps with 6 gliders, 8 blocks, 4 blinkers, 4 beehives, 1 boat, 1 ship, and 1 loaf.

(b) What does it mean to say "A toad is an oscillator with period 2."? [2 marks]

The toad is a stable (1), repeating pattern (1) that consists of two configurations (1). i.e. it will return to where it started in two time steps (1):

(c) Will a toad exist forever? [2 marks]

Maybe, it depends on what else is happening (2) – the toad is stable (1) but if it were to interact with something like a glider it would likely be destroyed (1).

(d) How does the glider gun refute Conway's conjecture? [4 marks]

Conway conjectured that there were no initial patterns that never stabilize (2). A gun is a stable pattern that periodically produces a spaceship (1)– as the stream of spaceships moves out from the source, the number of live cells grows indefinitely (1).

It was important to say what Conway's conjecture actually was and that the the number of live cells growing was the thing that contested it.